

Part No.	Date	Ver
HMS3014-YG	2020-11-30	1.0



DATA SHEET

Part NO. HMS3014-YG

MAKER			CUSTOMER		

NOTE: 3014 packaged 0.2W side view LED

Note: The information in this document is subject to change without notice.

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Features:

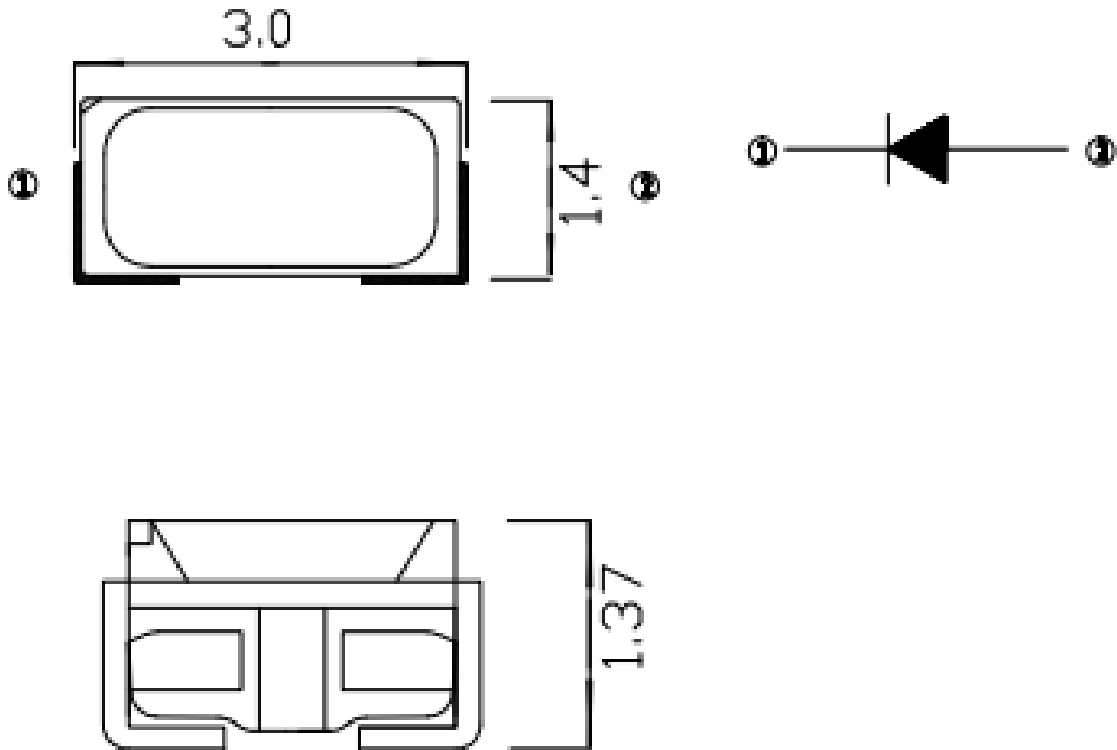
- Side view LED
- Yellow Green
- Thermally Enhanced Package Design
- High luminous flux output
- High current capability
- Compact Package Size
- Wide viewing angle
- Pb-free Reflow Soldering Application
- The product itself will remain within RoHS compliant version

Applications:

- Retrofits(replacement)
- General lighting
- Indoor & Outdoor sign board back light
- Architectural / Decorative lighting

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Package Dimension & soldering patterns:



- Notes:
- 1. All dimensions are in millimeters
 - 2. Tolerances unless dimensions $\pm 0.1\text{mm}$

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Absolute Maximum Ratings :

Parameter (Ta=25°C)	Symbol	Ratings	Unit
Power Dissipation	P _d	100	mW
Forward current	I _F	30	mA
Peak Forward Current	I _{FP}	60	mA
Reverse Voltage	V _R	5	V
Operating Temperature	T _{opr}	-40~+85	°C
Storage Temperature	T _{stg}	-40~+100	°C
Soldering Temperature	T _{slid}	Reflow Soldering: 260°C for 3 seconds	

- Pulse condition: Width≤100μs, Duty cycle≤1/10.
- LED’s properties might be different from suggested values like above and below tables if operation condition will be exceeded our parameter range. Care is to be taken that power dissipation does not exceed the absolute maximum rating of the product.
- All measurements were made under the standardized environment of Hivemico LED.

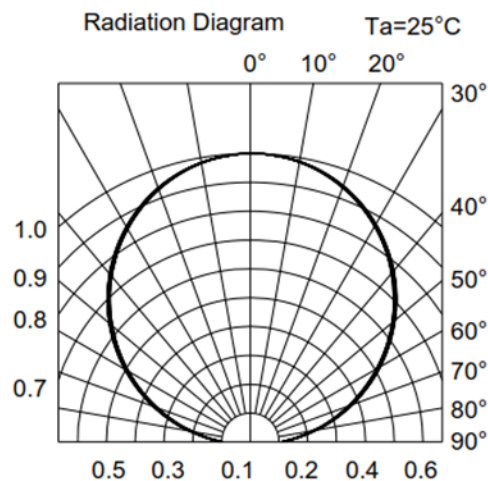
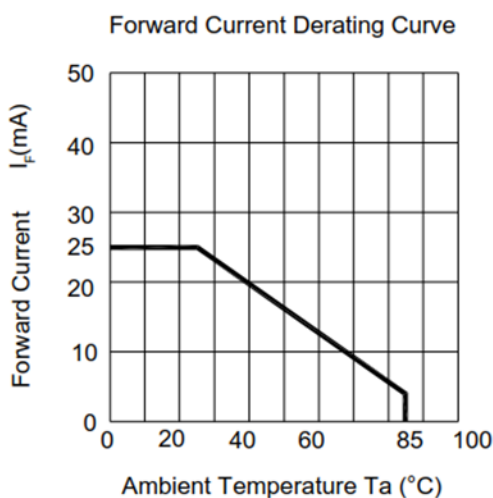
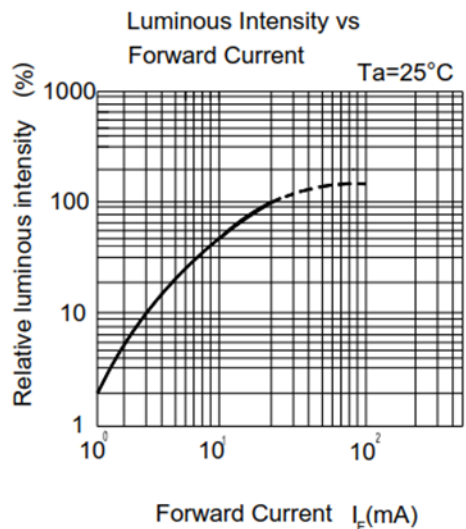
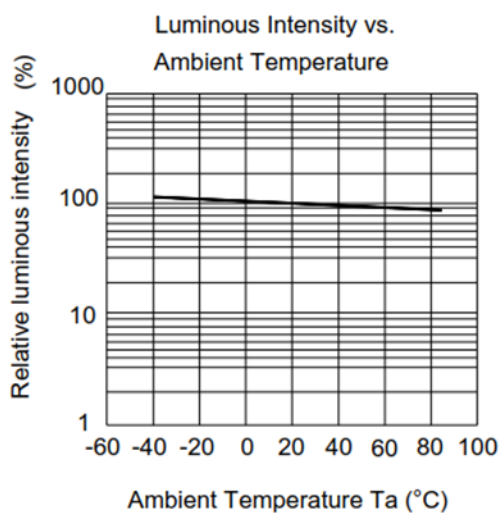
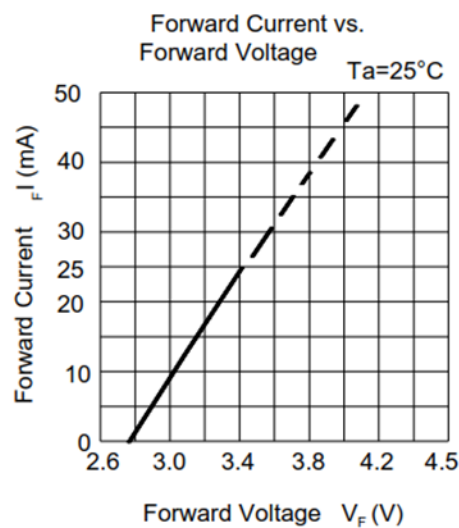
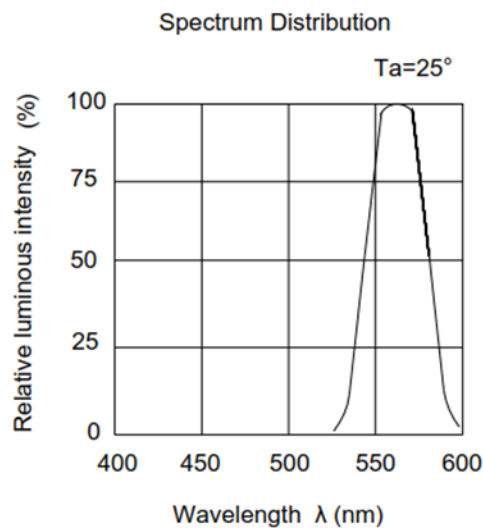
Electro-Optical Characteristics

Parameter (Ta=25°C)	Symbol	Condition	Min	Typ	Max.	Unit
Forward Voltage	V _f	I _f =20mA	2.5	--	2.9	V
Luminous Intensity	I _v	I _f =20mA	1400	--	2000	mcd
Reverse Current	I _R	V _R =5V	---	--	5	μA
Dominant Wave Length	λ _d	I _f =20mA	560	--	565	nm
Viewing Angle	2θ _{1/2}	I _f =20mA	---	120	---	deg

* Luminous flux is measured with an accuracy of ± 5%.
* Chromaticity coordinate bins are measured with an accuracy of ± 0.01.
* 2θ1/2 is the off-axis where the luminous intensity is 1/2 of the peak intensity.
* Rth j-sp is the thermal resistance from LED junction to solder point on MCPCB with electrical power

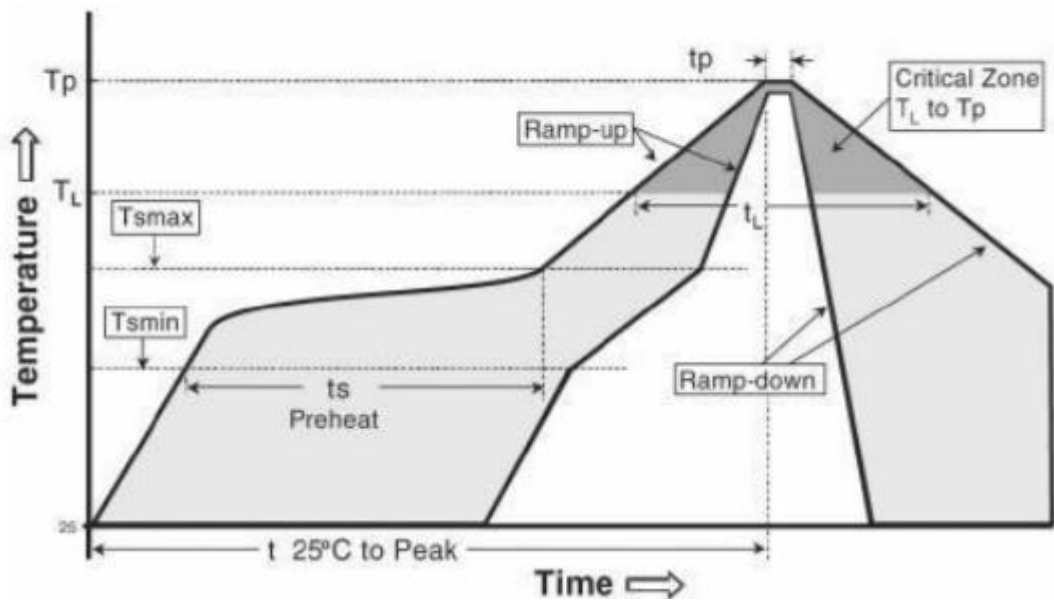
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Typical Electro-Optical Characteristics Curves



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Reflow Soldering Characteristics



Reflow soldering	
Temperature Min (Tmin) Temperature Max (Tmax) Time(ts)from (Tmin to Tmax)	150°C 200°C 60-120 seconds.
Ramp-up rate (TL to Tp)	3°C/seconds max
Liquidous temperature(TL) Time(tL) maintained above TL	217 °C 60-150 seconds
Peak package body temperature(Tp)	260°Cmax
Time (tp) within 5°Cof the specified classification temperature (Tc)	30 seconds max
Ramp-down rate (Tp to TL)	6°C/second max
Time 25 ° C to peak temperature	8 min max

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Reliability Test

Judgment criteria of failure for the reliability

Measuring items	Symbol	Measuring conditions	Judgement criteria for failure
Forward voltage	Vf (V)	I _f =20mA	Over Ux1.2
Reverse current	IR(μA)	V _R =5V	Over Ux1.2
Luminous Flux	Φv(lm)	I _f =20mA	Below SX0.5

Note:

1.U means the upper limit of specified characteristics. S means initial value.

2. Measurment shall be taken between 2 hours and after the test pieces have been returned to normal ambient conditions after completion of each test

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Classification	Test Item	Reference Standard	Test Conditions	Result
Endurance Test	Operation Life	MIL-STD-750:1026 MIL-STD-883:1005 JIS-C-7021 :B-1	Connect with a power If=20mA Ta=25°C ± 5°C Test time=1,000hrs	0/10
	High Temperature High Humidity Storage	MIL-STD-202:103B JIS-C-7021 :B-11	Ta=+60°C RH=90-95% Test time=240hrs	0/10
	High Temperature Storage	MIL-STD-883:1008 JIS-C-7021 :B-10	High Ta=+100°C Test time=1,000hrs	0/10
	Low Temperature Storage	JIS-C-7021 :B-12	Low Ta=-55°C Test time=1,000hrs	0/10
Environmental Test	Temperature Cycling	MIL-STD-202:107D MIL-STD-750:1051 MIL-STD-883:1010 JIS-C-7021 :A-4	-40°C ~ +25°C ~ +100°C ~ +25°C 30min 5min 50min 5min Test Time=500cycle	0/10
	Thermal Shock	MIL-STD-202:107D MIL-STD-750:1051 MIL-STD-883:1011	-40°C ~+100°C 15min 15min Test Time=10cycle	0/10
	Solder Resistance	MIL-STD-202:201A MIL-STD-750:2031 JIS-C-7021 :A-1	Preheating: 140°C-160°C,within 2 minutes. Operation heating: 260°C(Max.), within 10seconds. (Max.)	0/10

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CAUTIONS

1 . Application

The LEDs described here are intended to be used for ordinary electronic equipment (such as office equipment, communication equipment and household applications).Consult Sales in advance for information on applications in which exceptional reliability is required, particularly when the failure or malfunction of the LEDs may directly jeopardize life or health (such as in aviation, transportation, traffic control equipment, medical and life support systems and safety devices).

2. Storage

The storage ambient for the LEDs should not exceed 30°C temperature or 70% relative humidity. It is recommended that LEDs out of their original packaging are IR-reflowed within one week. For extended storage out of their original packaging, it is recommended that the LEDs be stored in a sealed container with appropriate desiccant, or in a desiccators with nitrogen ambient. LEDs stored out of their original packaging for more than a week should be baked at about 60 °C for at least 24 hours before solder assembly.

3. Cleaning

Do not use unspecified chemical liquid to clean LED they could harm the package. If clean is necessary, immerse the LED in ethyl alcohol or in isopropyl alcohol at normal temperature for less one minute.

4. ESD (Electrostatic Discharge)

Static Electricity or power surge will damage the LED. Here is some Suggestions to prevent ESD damage: (1)Use of a conductive wrist band or anti-electrostatic glove when handling these LEDs. (2)All devices, equipment, and machinery must be properly grounded. (3) Work tables, storage racks, etc. should be properly grounded. (4) Use ion blower to neutralize the static charge which might have built up on surface of the LED’s plastic lens as a result of friction between LEDs during storage and handling.

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ESD-damaged LEDs will exhibit abnormal characteristics such as high reverse leakage current, low forward voltage, or “ no lightup ” at low currents. To verify for ESD damage, check for “ lightup ” and Vf of the suspect LEDs at low currents. The Vf of “ good ” LEDs should be >2.0V@0.1mA for InGaN product and >1.4V@0.1mA for AlInGaP product.

5. Drive Method

An LED is a current-operated device. In order to ensure intensity uniformity on multiple LEDs connected in parallel in an application, it is recommended that a current limiting resistor be incorporated in the drive circuit, in series with each LED . Otherwise ,The brightness of each LED might appear different due to the differences in the I-V characteristics of those LEDs.

6. Soldering

1 Manual Of Soldering: The temperature of the iron tip should not be higher than 300°C(572°F) and Soldering within 3 seconds per solder-land is to be observed.

2 Reflow Soldering Preheating : 140°C~160°C ± 5°C,within 2 minutes.
Operation heating : 260°C(Max.) within 10 seconds.(Max) Gradual Cooling (Avoid quenching).

3 DIP soldering (Wave Soldering) : Preheating : 120°C~150°C,within 120~180 sec. Operation heating : 245°C ± 5°C within 5 sec.260°C (Max) Gradual Cooling (Avoid quenching).